

6. Analyze and understand the role of advanced welding techniques towards industry and societal impact.

Text Books

1. R.S., Parmar, Welding Engineering and Technology, Khanna Publishers, 2002
2. Sindo Kou, Welding Metallurgy, Wiley-Interscience, John Wiley & Sons
Nadkari, S. V., Modern Arc Welding Technology, Oxford & India Book House Pvt. Ltd. (2005)

Reference Books

1. ASM Handbook – welding, brazing and soldering, vol 6,
2. Howard B Cary, Modern welding technology, 6th Ed., Prentice Hall USA, 2004

Evaluation Scheme

S. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	45
3	Sessionals (Assignments/Projects/Tutorials/Quizzes)	30

UME744 : Hydrogen Fuel Cell Technology

L	T	P	Cr
3	0	0	3.0

Course Objective: To provide thorough understanding of performance characteristics of fuel cell power plant and its components. Outline the performance and design characteristics and operating issues for various fuel cells. Discuss the design philosophy and challenges to make this technology economically feasible. The design and analysis emphasis will be on the thermodynamics and electrochemistry. Thus, at the successful end of the course, the students will have sufficient knowledge for working in a fuel cell industry or R&D organization.

Syllabus

Introduction: Hydrogen and its properties – physical and chemical; applications of hydrogen as a source of energy; availability of hydrogen, production and storage of hydrogen.

Fuel cell technology: History of fuel cells, advantages and disadvantages of fuel cells, applications of fuel cells, principle of operation, fuel cell classification, characteristics, fuel cell stacking, membrane-electrode-assembly (MEA), flow field plates, humidifiers and associated controls.

Fuel cell performance: the role of Gibbs Free Energy, ideal performance, cell energy balance, fuel cell efficiency, actual performance, performance variables, polarization curves, power characteristics, temperature and pressure effects, hydrogen purity effect, stoichiometric effect, humidity effects.

Fuel cell engine systems: system descriptions, air system, fuel storage system, fuel delivery system, humidification system, stack cooling system and controls, leak detection system, HVAC system.

Fuel cell hybrid electric vehicles: hybrid electric vehicles – series hybrid and parallel hybrid; major components of hybrid vehicles, electric drive motors, auxiliary power units, generators, energy storage systems, regenerative braking, control system.

Fuel cell engine safety: hydrogen leaks and detection, hydrogen fires, low and high temperature hazards, electrical shock, de-ionizing resin, purple K dry chemical fire retardant.

Course Learning Objectives (CLO)

The students will be able to:

1. design and perform analysis of fuel cell.
2. evaluate the performance behaviour, operational issues and challenges for all major types of fuel cells.
3. identify, formulate, and solve problems related to fuel cell technology keeping in mind economic viability.
4. compare fuel cell powered vehicles with vehicles running on I.C. engine.

Text Books

1. Ohayre, Ryan P. Fuel cell fundamentals, John Wiley and Sons, 2006.
2. Li. Xianguo, Principles of fuel cells, Taylor and Francis, 2006.
3. Sorensen, Bent. Hydrogen and fuel cells: emerging technologies and applications, Elsevier, 2012.
4. Bagotsky, Vladimir S. Fuel cells: problems and solutions, Willey, 2008